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Faraday cup for measuring the electron beams of TWT guns

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A B S T R A C T

A compact Faraday cup reported in this paper is designed to investigate the current distributions of the electron beams of Traveling Wave Tube's guns. It consists of a 0.06 mm thick molybdenum aperture plate and a copper shield with a graphite collector inside. There are plates with different laser-cut aperture holes that were 100 μm, 50 μm, 20 μm and 10 μm in diameter for measuring electron beams with different sizes. The thermal analysis of the Faraday cup with pulse beam heating was performed and discussed in this paper. The pulse test shows that this device has fast response and small dissipation. A 0.58 beam perveance (μP) electron gun with expected minimum beam radius 1.0 mm was measured with the Faraday cup and the three-dimensional current density distribution and beam envelope were presented. The experiment results show that the design is reasonable for measuring the electron beam with a high resolution.

1. Introduction

The performance of traveling wave tube (TWT) depends critically on the parameters of the electron beam that drives the device [1]. Generally the electrons emitted from the spherical cathode of a Pierce-type gun are brought to a focus in the vicinity of the anode aperture before injection into the radio frequency (RF) circuit [2]. Only the use of an electron beam of optimum design will lead to an effective beam–wave interaction and satisfactory tube performance. However, sometimes the beam transmission efficiency is not as high as that was designed. Possibly the parameters of the real beam may be quite different from the results of calculation or simulation based on computer simulations of the electron optics, because of thermal effects, cathode roughness and gun aberrations, etc. Therefore it is necessary to investigate the properties of the electron beam. A Faraday cup becomes a nature choice for measuring low-energy electron beam of TWT gun because it is a useful, simple and compact device routinely used to determine the current distribution of the electron beam [3,4].

There are many different types of Faraday cup. Traditional Faraday cups have been widely used in many applications, such as in plasmas [5], accelerator beam diagnostics [6–8], ion beams [9,10], etc. However, all these Faraday cups are not suitable to scan the electron beam repeatedly to obtain the current distribution as use in the measurements of the TWT guns.

This paper describes a compact Faraday cup designed to determine the spatial distribution of the electron beams from acceleration voltage $U \leq 15$ kV guns of TWTs. The Faraday cup consists of a 0.06 mm thick molybdenum aperture plate and a copper shield with a graphite collector inside. The aperture plate with a pinhole on it can sample the electron beam on a square scanning area centered about the beam and then the current density distribution of the beam is constructed. Furthermore, after that series of three-dimensional profiles at successive of transverse planes are established, the beam envelope, minimum radius and its axial location are available. First, the design of the Faraday cup is discussed. Then, a TWT gun measurement results with the Faraday cup are presented.

2. Design of the Faraday cup

An aperture plate and a collector are mounted together as a Faraday cup assembly to scan the electron beam in an ultra-high vacuum (UHV) chamber as described in ref. [4]. During a pulse, the electron beam strikes the aperture plate, and only a fraction of the beam is allowed to go through the aperture hole to the collector. In order to reduce the capture by the wall when the electron travels though the aperture, the plate should be made as thin as possible. The collector must be able to capture the electrons and reduce the secondary electrons with fast response.

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2.1. Basic design

Considering the repetitive scanning mission of the Faraday cup, the electron beam might destroy it after all. So its components should have replacements. Moreover, the Faraday cup should be assembled easily.

A schematic of the design is shown in Fig. 1. The signal in the Faraday cup can be connected to the feedthrough on the UHV chamber by a connector on which the copper outer cylindrical shield can be fixed by screw thread. There are a special design graphite collector inside the outer shield and a ceramic insulator between them. A copper cap with punch hole is placed on the head of the outer shield by screw. Between them there is a 0.06 mm thick molybdenum plate with a pinhole. The basic design makes sure the Faraday cup can be assembled easily and quickly.

To stop 15 keV electrons completely would theoretically require about 3.8×10^{-4} g/cm^2 of graphite [11]. In our case the thickness of the bottom wall and side wall of the graphite collector are about 2.0 mm and 0.45 mm, giving about 0.5 g/cm^2 and 0.1 g/cm^2, respectively, guaranteeing fully zero escape of electrons through the walls of the collector. Moreover the graphite material can reduce the secondary electron emission effectively. We made molybdenum plates with different laser-cut aperture holes that were 100 μm, 50 μm, 20 μm and 10 μm in diameter to measure beams with different sizes.

2.2. Thermal analysis and pulse test

The cathode emission of the TWT gun would be more stable with a longer pulse. However, the temperature rises to a high value in a very thin surface layer of the aperture plate during a pulse, and then the plate may melt. That will completely destroy the Faraday cup. It is assumed that all intercepted beam power turns into heat power and is stored by diffusion into the plate, then the pulse temperature rise as a function of peak intercepted power density, P, beam radius, b, and pulse width, τ, is [12]: Fig. 2

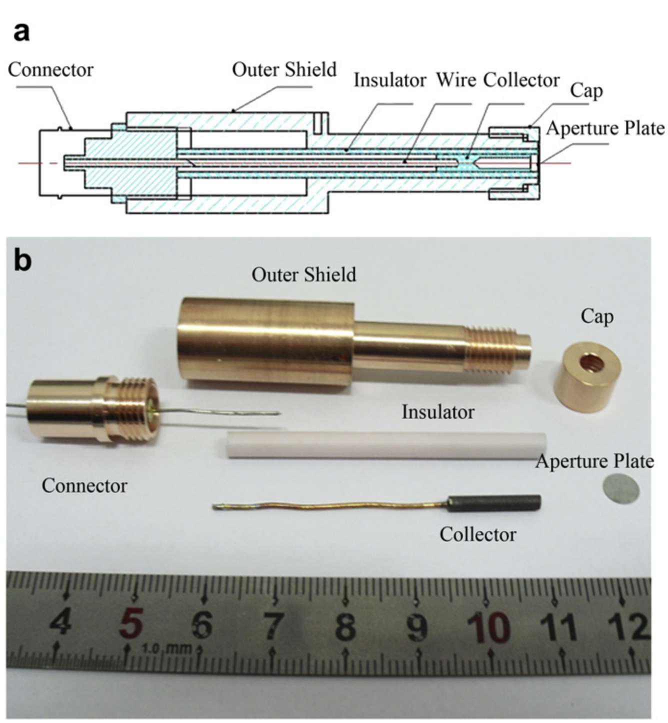

Fig. 1. View of the Faraday cup. The maximum outer diameter of the outer shield is 14 mm. The outer diameters of the ceramic insulator, aperture plate and graphite collector are 4.0 mm, 6.0 mm and 2.4 mm, respectively. (a) Schematic of the Faraday cup. (b) Components.

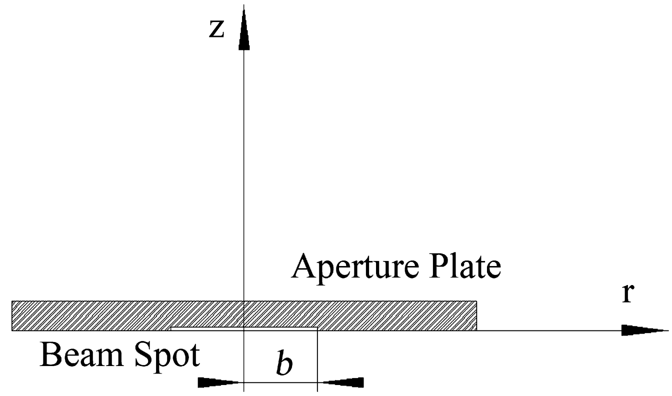

Fig. 2. Aperture plate with beam spot.

$$T(r,z,\tau) = \frac{Pb}{2K}\int_0^\infty \beta^{-1}J_0(\beta r)J_1(\beta b)\left[\exp(-\beta z)erfc\left(\frac{z}{2\sqrt{a\tau}}-\beta\sqrt{a\tau}\right) - \exp(\beta z)erfc\left(\frac{z}{2\sqrt{a\tau}}+\beta\sqrt{a\tau}\right)\right]d\beta \tag{1}$$

$$erfc(\eta) = 1-\frac{2}{\sqrt{\pi}}\int_0^\eta \exp\left(-\xi^2\right)d\xi \tag{2}$$

$$ierfc(\eta) = \int_\eta^\infty erfc(\xi)d\xi \tag{3}$$

$J_0(r)$ and $J_1(r)$ are the first kind Bessel functions. Due to the short pulse which width is not longer than a few milliseconds, the problem can be treated as a surface phenomenon. So at the center of the beam spot on the plate where is $r = z = 0$, the maximum pulse temperature rise is

$$\Delta T = \frac{2P\sqrt{\tau}}{\sqrt{\pi K\rho c}}\left[1-\sqrt{\pi}ierfc\left(\frac{b}{\sqrt{4a\tau}}\right)\right] \tag{4}$$

here K, c, ρ and $a = K/(\rho c)$ are the thermal conductivity, specific heat, density, and thermal diffusivity, respectively Table 1.

The electron beam from acceleration voltage $U = 3.5$ kV, beam current $I = 120$ mA gun of TWT was measured. The expected size of the beam is about 2.0 mm and the peak beam power density is about 133.7 MW/m^2. The plate would melt in about 129 ms with continuous beam strike. It was assumed that the pulse temperature riseΔT did not exceed 1000 °C, so the pulse width was about 15 ms. For normal operating, 30 μs pulse was employed in the experiment and that would cause 44 °C temperature rise in theory and 42 °C in simulation using ANSYS. In addition, considering the heat cumulating effect, suitable pulse frequency should be employed as well. If the initial temperature of the Faraday cup is assumed to be 100 °C,

Table 1
Properties of molybdenum plate.

Material	K(W·m^{-1}·K^{-1})	ρ (kg·m^3)	c (J·kg^{-1}·K^{-1})
Mo	142	10230	242

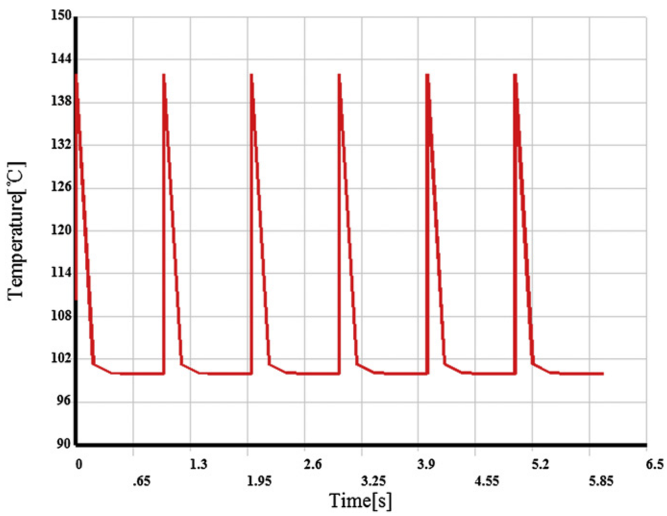

Fig. 3. Transient maximum temperature of the Faraday cup.

the steady-state temperature with the average power is 100.3 °C, and the maximum temperature on it changing over time with 1.0 Hz pulse is shown in Fig. 3. It seems that the temperature will rush to a peak value142 °C during each pulse and then fall to the steady-state value. Short pulse with long period ensures the plate would not melt due to the high current density. In addition, the Faraday cup has favorable heat-sinking capability for the good thermal conductivity of the copper shield.

The response of this device tested by a digital oscilloscope (Agilent Model InfiniiVision DSO9254A) is shown schematically in Fig. 4. A typical pulse response trace is shown in Fig. 4 (a) where the upper trace is the input pulse and the lower trace is the output of the cup. The risetime of this device is shown to be a few nanoseconds. The continuous pulses response of the cup is shown in Fig. 4 (b) where the output trace is coincidence with the input trace. And it seems that there is dissipation during the single pulse response in Fig. 4 (a), but the Fig. 4 (b) shows that the dissipation is ignorable.

3. Experiment results

The Faraday cup mounted on an X, Y, Z platform in the UHV chamber of a beam analyzer is shown in Fig. 5. The vacuum pressure must be better than 5×10^{-6} Pa. The TWT gun is mounted on the flange in the chamber and triggered by a pulse modulator controlled by computer. The electron beam hits the aperture plate of the Faraday cup while the pinhole allows only a small part of the beam through to the collector. The electrons captured by the collector would raise a current pulse signal I_n. This current is proportional to the local current density.

$$I_n = J_n \pi r_h^2 \tag{5}$$

here J_n is the local current density of the beam and r_h is the radius of the pinhole. The Faraday cup is confined to move to each node of an imaginary grid imposed on a square scanning area centered about the beam. When all the J_n at each node is taken, three-dimensional

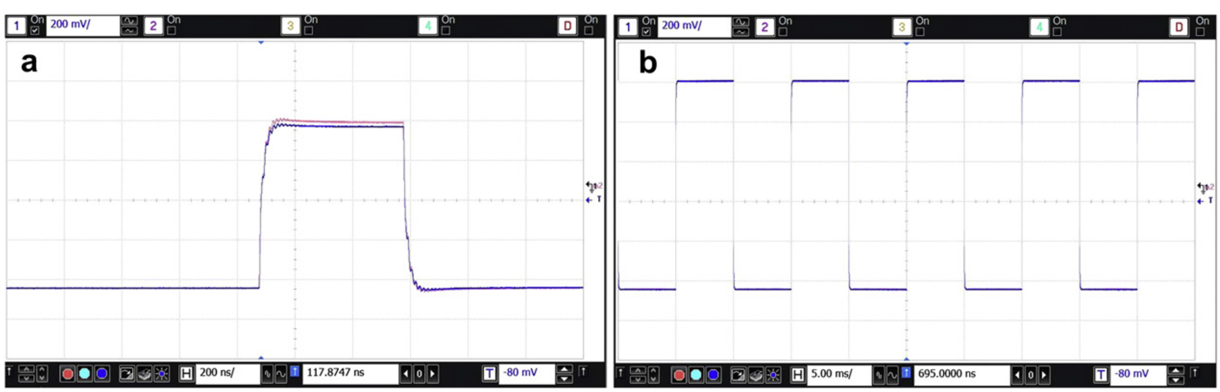

Fig. 4. Pulse response of the Faraday cup. (a) 500 ns single pulse response. (b) Continuous pulses response.

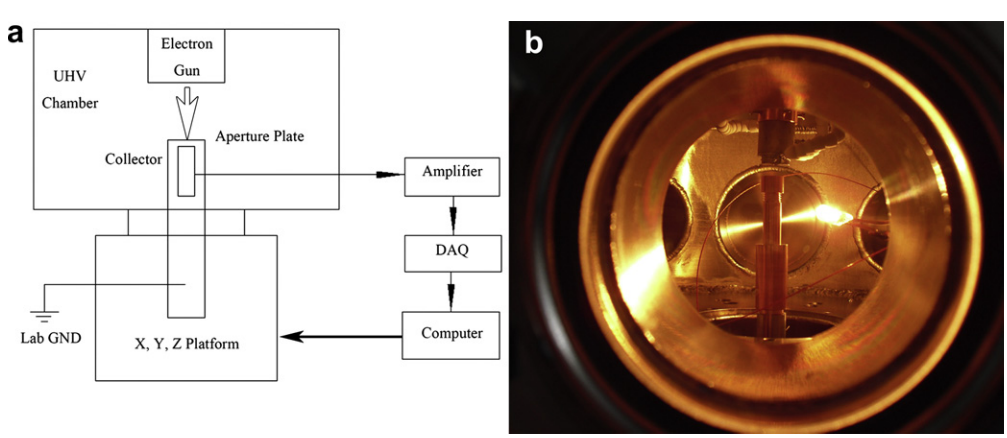

Fig. 5. Experimental setup for measuring the electron beam. (a) Schematic of the beam analyzer. (b) Faraday cup in the UHV chamber.

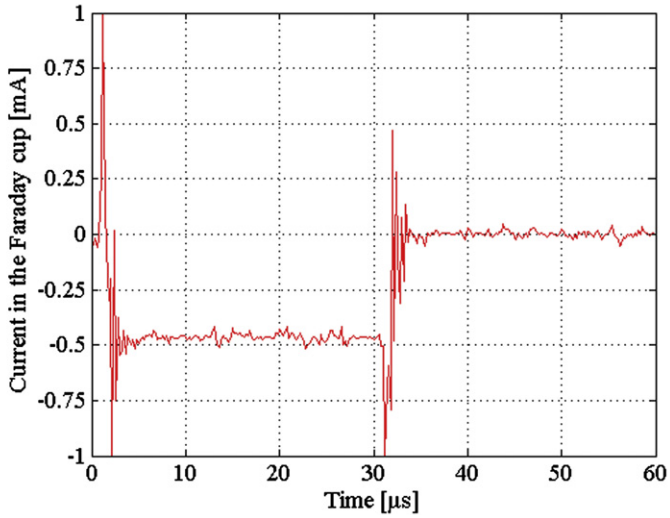

Fig. 6. Typical current signal in the Faraday cup.

current density distribution of the beam will be available. Each axis of the platform could provide a resolution as low as 0.5 μm. Naturally the resolutions of the beam profiles depend upon the size of the pinhole and the spacing of the nodes.

In this experiment it has demonstrated a resolution of 0.1 mm with a pinhole 0.1 mm in diameter and 0.1 mm spacing.

A typical current signal in the Faraday cup is shown in Fig. 6. The pulse width is 30 μs with 5M/s sampling rate. The testing gun was controlled by modulating anode, and the voltage range over which the modulating anode must be varied was large, which means that there was sharply variational electric field in the UHV chamber. Due to the induced capacitance and inductance of the circuit, there were oscillations during the pulse rising and trailing edge. However, only the stable part of the pulse would be used where the cathode emission was considered to be stable. So the oscillation and transition phenomenon of the pulse had marginal impact on the results.

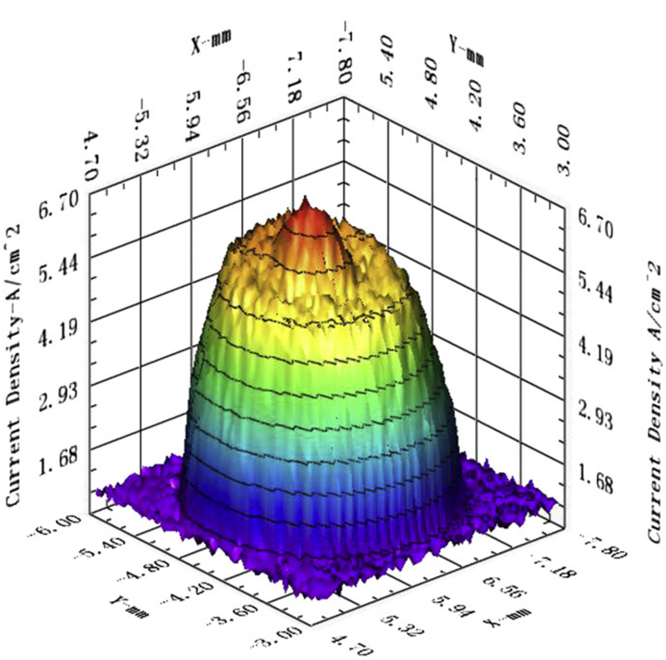

Fig. 7. Three-dimensional beam current density distribution.

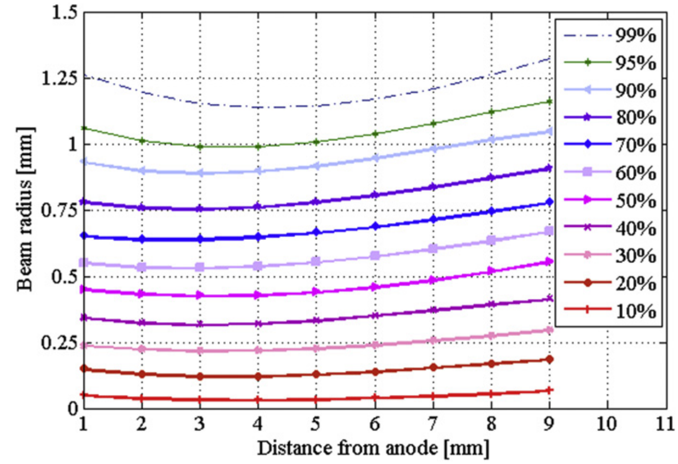

Fig. 8. Beam envelopes of different fractions of the total beam current.

A typical three-dimensional beam current density distribution taken 3 mm in front of the anode is illustrated in Fig. 7. Apparently it is a nonlaminar beam with a peak in the center of the beam current density. It is thought that the peak resulted from ion effects in the center of the beam. The ions in the gun area being accelerated toward the cathode, along with sputtered cathode material that becomes ionized, neutralized the electron space charge cloud near the center of the cathode so that additional electrons leaved the cathode. Moreover the cause which leaded to the inclined beam density distribution might be the installation error of the electrodes. Yet at the same time the thermal effects, gun aberration and patchy emission, etc. might also affect the result.

When the beam diameters of series of three-dimensional profiles at successive of transverse planes are established, the beam envelope is determined. Fig. 8 shows the beam envelopes of different fractions of the total beam current. That provides such information as minimum beam radius and its axial location. The beam waist radius is about 1.0 mm (95% of the total current), which is equal to the predicted value.

The aperture plate of the Faraday cup could be replaced by another one with different size of aperture hole, and the pulse could be regulated by the computer, then this device is applicable to other guns with different beam perveances and sizes.

4. Conclusion

A compact Faraday cup has been designed to investigate the spatial properties of the electron beams of TWT guns. The Faraday cup consists of a thin molybdenum aperture plate and a copper shield with a graphite collector inside. The pulse test shows that this device has fast response and small dissipation. It has demonstrated a high resolution at the beam current density distribution measurement. The experiment results indicate that the Faraday cup can satisfy the measure requirement with good performance.

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